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### Insect monitoring system

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#### Abstract

An insect monitoring system for use in agricultural fields is provided. The monitoring system has a chemical lure to attract insects to the monitoring system, and an adhesive material on which insects drawn to the lure can be trapped for later evaluation. The monitoring system also has a three-piece canopy made of two opposing sides and a cover. The parallel sides are roughly triangular and the cover sits above, and is attached to, the sides. The adhesive trap is held vertically between the two opposing sides of the canopy. The lure is held horizontally above the adhesive trap by being inserted through slits in the cover. The canopy can be assembled from a flat sheet of a prepatterned rigid material.

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**Field of Classification Search****CPC:**      A01M (1/026); A01M (1/14); A01M (1/16); A01M (1/145); A01M (1/165)**References Cited****U.S. PATENT DOCUMENTS**

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*Primary Examiner:* Wang; Michael H*Attorney, Agent or Firm:* Rimon Law, P.C.**Background/Summary**

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part of U.S. design application Ser. No. 29/901,987 filed on Sep. 7, 2023, the disclosure of which is incorporated by reference herein.

**BACKGROUND OF THE INVENTION****Field of the Invention**

(1) The invention relates generally to the field of agriculture and more particularly to monitoring destructive pests.

**Description of the Prior Art**

(2) A large variety of insect pests pose threats to commercial agricultural crops. In order to control

these pests, it is typically necessary to monitor the agricultural fields for signs of infestation. One component of such monitoring is the use of distributed insect traps. Periodically, these traps need to be inspected to count the numbers of different insect species that have been collected to determine whether the field is, or is becoming, infested, and with what insects, in order to decide how and when to treat the field.

## SUMMARY

(3) The present invention is directed to an insect monitoring system, a method of assembling the insect monitoring system, and a kit containing the parts necessary to assemble the insect monitoring system. An exemplary insect monitoring system disclosed herein comprises a canopy, a lure, and an adhesive trap. The lure includes an insect attractant to draw insects to the monitoring system where they get caught on the adhesive trap. The canopy protects the lure and adhesive trap against the elements.

(4) More specifically, an exemplary insect monitoring system according to the present invention comprises a canopy including two parallel opposing vertical sides and a cover spanning between the opposing sides, the cover having two portions that are mirror images of one another subtend an angle. The opposing vertical sides and the cover of the canopy can comprise, in some embodiments a single folded sheet of a rigid material. The cover optionally includes a third portion between the two portions, the third portion being disposed horizontally and parallel to the lure. The two portions of the cover each include a horizontal slit sized to receive the lure therethrough.

(5) The opposing vertical sides of the canopy optionally include both an angled slit and a hole, where each angled slit extends from a bottom edge of the respective vertical side to the hole. In some of these embodiments each opposing vertical side includes a notch aligned with the hole and extending from the bottom edge towards the hole. The arrangement of angled slits, holes, and notches allows an adhesive trap to be suspended from the vertical sides, and removed later for inspection.

(6) The exemplary insect monitoring system optionally can also comprise a means for hanging the insect monitoring system that attaches to the canopy. For example, the means for hanging can comprise a wire. In some embodiments the ends of the wire are disposed through holes in the canopy.

(7) In an exemplary method for assembling an insect monitoring system, a sheet of a rigid material having holes and slits defined therethrough is received. Then, the sheet is folded to form a cover defined by two mirror image portions, each portion including two slits. Next, each of two parallel opposing vertical sides are attached to the cover, each side including two tabs. Each side is attached by inserting a first of the two tabs through a slit of the two slits of one portion of the cover and inserting a second of the two tabs through a corresponding slit of the other portion of the cover. In various embodiments, folding the sheet to form the cover includes folding the sheet to form three portions, wherein the three portions include the two mirror image portions and a third portion in the shape of a rectangle disposed between the other two portions. In some of these embodiments, the parallel opposing vertical sides are joined to the short sides of the third portion, and the sheet is folded along the two short sides before each of the two parallel opposing vertical sides are attached to the cover.

(8) Once the canopy has been completed in this way, the lure, including a pouch having an insect attractant disposed therein, is inserted through a horizontal slit defined in each of the two portions of the cover. The adhesive trap, including a card having an adhesive disposed thereon, is then attached to the opposing vertical sides of the canopy such that the adhesive trap is held vertically beneath the lure. In some embodiments, each opposing vertical side includes an angled slit extending from a bottom edge thereof to a hole therethrough, the opposing vertical sides are mirror images of each other, and the adhesive trap includes two slits therethrough. Here, attaching the adhesive trap to the opposing vertical sides includes inserting the adhesive trap into the two angled slits.

(9) An exemplary kit of the present invention comprises a packaged lure including a pouch having an insect attractant disposed therein, a packaged adhesive trap including a card having an adhesive disposed thereon, and a precut sheet of a rigid material cut to define two portions, the portions being mirror images of each other, each portion including two tab-receiving slits and one horizontal slit. The kit optionally further comprises a means for hanging an insect monitoring system assembled from the precut sheet, lure, and adhesive trap.

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## Description

### BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a top perspective view of an exemplary insect monitoring system according to various embodiments.

(2) FIG. 2 is a bottom perspective view of the insect monitoring system of FIG. 1.

(3) FIG. 3 is an exploded view of the insect monitoring system of FIG. 1.

(4) FIG. 4 is a top view of an exemplary canopy before assembly, according to various embodiments.

(5) FIG. 5 shows exemplary steps of the assembly of an insect monitoring system according to various embodiments.

(6) FIGS. 6A-6C show another exemplary step of the assembly of the insect monitoring system according to various embodiments.

### DETAILED DESCRIPTION

(7) The present disclosure is directed to an insect monitoring system that is simple to deploy, for example, in orchards and that includes a lure to attract insects and an adhesive material on which insects drawn to the lure can be trapped for later evaluation. In various embodiments, the insect monitoring system ships flat, is simple to assemble, and simple to replace depleted lures and used adhesive material.

(8) FIG. 1 provides a prospective view of an exemplary insect monitoring system **100** according to various embodiments. The monitoring system **100** comprises a canopy **105**, an adhesive trap **110**, a lure **115**, and means **120** for hanging the monitoring system **100**, such as from a tree.

(9) The canopy **105** is preferably made from a sheet of a rigid material such as plastic, card stock, or cardboard. An exemplary plastic comprises polypropylene. For example, 2 mm, 3 mm, or 4 mm thick corrugated plastic can be used. The sheet is folded, such as along scored lines, to form a cover **125** and two parallel opposing vertical sides **130**, **135**. The cover **125** of the canopy **105** includes two portions **140**, **145** that subtend an angle,  $(180-2\alpha)$ , and optionally includes a flat third portion **150** joining the two portions **140**, **145**, as shown. In embodiments without the third portion **150**, the two portions **140**, **145** meet along a line. The angle,  $\alpha$ , is the angle each portion **140**, **145** makes with the horizontal, which can be between  $25^\circ$  and  $35^\circ$ , for instance  $30^\circ$ .

(10) Advantageously, the sheet of rigid material can be patterned, for instance, with score lines, cut-out areas, and with holes, slits, and notches, such as by die cutting. Thus, multiple canopies **105** can be shipped as a stack of sheets together with a set of lures **115**, also packed as multiple individually-wrapped stacked sheets, and a stack of adhesive traps **110**, similarly packed as multiple individually-wrapped stacked sheets. In the field, a worker can rapidly fold-up a canopy **105**, insert a lure **115**, add means **120** for hanging, and install an adhesive trap **110**, then hang the system **100**.

(11) The two sides **130**, **135** of the canopy **105** are joined to the two portions **140**, **145** of the cover **125**. In some embodiments, as illustrated in FIG. 1, the sides **130**, **135** include tabs **155**, the two portions **140**, **145** include slits **160**, and the tabs **155** fit through the slits **160** to hold the sides **130**, **135** in place. In other embodiments the sides **130**, **135** can be attached to the two portions **140**, **145** of the cover **125** with adhesive strips, for instance.

(12) The two portions **140**, **145** of the cover **125** each include, in various embodiments, a horizontal slit **165** sized such that the lure **115** can be inserted through both slits **165**. In this way the lure **115** is retained in a horizontal orientation by the cover **125**. It can be seen that the lure **115** is held by friction and can be readily removed and replaced.

(13) The two sides **130**, **135** of the canopy **105** are configured to retain the adhesive trap **110** in a vertical orientation between them, and beneath the lure **115**. As illustrated, in some embodiments each of the sides **130**, **135** includes an angled slit **170** extending from a bottom edge to a hole **175**. In some of these embodiments each of the sides **130**, **135** further includes a notch **180** aligned beneath the hole **175** and extending towards the hole **175** from the bottom edge.

(14) The adhesive trap **110** can be in the form of a paper or plastic card having an adhesive layer on both sides. In embodiments that include the angled slit **170** and hole **175**, the adhesive trap **110** includes two vertical slits **185** spaced apart by the same distance that separates the two sides **130**, **135**. In these embodiments, a top edge of the trap **110** can be inserted through the two angled slits **170** until the top edge is within the hole **175**. The vertical slits **185** allow the adhesive trap **110** to then be rotated to a vertical orientation. As with the lure **115**, the adhesive trap **110** can be readily removed and replaced. This is further described with respect to FIGS. 6A-6C.

(15) In other embodiments the adhesive trap **110** attaches to the two sides **130**, **135** with adhesive strips. In still other embodiments, the two sides **130**, **135** each includes a long vertical slit, the adhesive trap **110** is pushed up through the two vertical slits, and the portions of the adhesive trap **110** that extend outside of the canopy **105** are taped to the respective sides **130**, **135**. In some of these embodiments the portions of the adhesive trap **110** that extend outside of the canopy **105** are folded to lay flat against the respective sides **130**, **135** then taped in place.

(16) The means **120** for hanging the monitoring system **100** attaches to the canopy **105**. In the illustrated embodiment the means **120** comprises a wire that is fed through holes **190** in the cover **125** of the canopy **105**. Optionally, the wire also is fed through a hole (not shown) in the lure **115** that is aligned beneath one of the holes **190**. This serves to further secure the lure **115**. The means **120** can include a top loop **195** that can be placed over a branch or a nail or some other protrusion from which the monitoring system **100** can be suspended. The means **120** can include, instead of a top loop **195**, a hook such as that of a clothes hanger.

(17) FIG. 2 shows a bottom perspective view of the insect monitoring system **100**. The bottom view shows the optional hole **200** through the lure **115** through which the means **120** is disposed in this embodiment. In the given embodiment the wire of the means **120** is continuous from one hole **190** to the other, but in other embodiments the wire is not continuous in this way, and has two ends that each extend through a respective hole **190** and are bent under the cover **125**. The means **120** can also comprise a rod with a threaded end that extends through a single hole **190** in the cover **125** and is secured by a washer and nut.

(18) The lure **115**, in some embodiments, comprises a pouch made from barrier layer, semipermeable layers, and peelable layers with the attractant disposed between the outside barrier layer and the semipermeable layers. In order to monitor for Spotted Wing Drosophila (SWD), for example, suitable attractants include a formulated combination of acetic acid ethanol, 3-hydroxy-2-butanone, 3-(methylthio)-1-propanol, and Klucel H IND.

(19) The adhesive trap, in various embodiments, comprises 20 mil thick low-density polyethylene (LDPE) substrate coated with a hot melt adhesive.

(20) FIG. 3 shows an exploded view of the insect monitoring system **100**.

(21) FIG. 4 shows an exemplary flat sheet **400** that can be folded to form the canopy **105**. In embodiments without the third portion **150**, the vertical sides **130**, **135** can be separate pieces, each can include two or more tabs **155** and the two portions **140**, **145** of the cover **125** include appropriate corresponding slits **160**.

(22) FIG. 5 shows steps of the assembly of the insect monitoring system **100** following forming of the canopy **105**. First, the lure **115** is inserted through the horizontal slits **165** defined in each of the

two portions **140**, **145** of the cover **125**. In some embodiments, this includes aligning a hole **200** through the lure **115** with a hole **190** in the cover **125**.

(23) Next, in the illustrated embodiment, means **120** for hanging are attached. In the illustrated embodiment the means **120** comprises a single wire **500** formed into a loop **195** at the middle of the length of the wire **500**, with the two halves of the wire **500** twisted around each other to form a length **510** with some rigidity. The remaining ends **520**, **530** of the wire **500** are bent perpendicular to the length **510** and pointing away from each other, and then bent again to be parallel to the length **510**. In these embodiments the free ends **540**, **550** of the wire **500** are inserted completely through holes **190** in the cover **125**, bent towards one another, and optionally further twisted together.

(24) Other means **120** for hanging the insect monitoring system **100** include looping a string through the holes **190**, and taping or gluing a string to the top of the canopy **105**. The sheet **400** can also be fabricated with a tab having a hole disposed at the center of the third portion **150** that can be folded up so that a wire or thread can be threaded through the hole.

(25) FIGS. **6A-6C** illustrate an exemplary step for attaching the adhesive trap **110** to the opposing vertical sides **130**, **135**. As shown in FIG. **6A**, in some embodiments of the adhesive trap **110**, a top of the adhesive trap **110** comprises a strip **600** that runs the length of the adhesive trap **110**.

Attaching the adhesive trap **110** to the opposing vertical sides **130**, **135** comprises, in these embodiments, bending the strip **600** relative to the remainder of the adhesive trap **110** to match the angle of the slit **170**. Then, as shown in FIG. **6B**, the strip **600** is pushed up along the slit **170**, and then as illustrated by FIG. **6C** the strip **600** fits into the holes **175** and the vertical slits **185** allow the adhesive trap **110** to be disposed vertically. As noted above, in other embodiments the adhesive trap **110** does not include the strip **600** and is installed similarly.

(26) The present invention also includes a kit for an insect monitoring system. Such a kit comprises a packaged lure, a packaged adhesive trap, and a precut sheet of a rigid material. The kit optionally further comprises a means for hanging an insect monitoring system assembled from the precut sheet, lure, and adhesive trap. The lure is packaged to keep the insect attractant from dissipating until the monitoring system is deployed, and can comprise a foil pouch, for instance. The adhesive trap is likewise packaged to keep the adhesive from becoming contaminated before the monitoring system is deployed. Packaging for the adhesive trap can include peel-off paper backing and/or a foil pouch.

(27) As noted previously, precut sheets can be stacked for shipment. As used herein, a kit can be an assembly of one packaged lure, one packaged adhesive trap, and one precut sheet of a rigid material packed together, but can also comprise a package of precut sheets with a number of packaged lures and a number of packaged adhesive traps. In various embodiments a kit can include more lures, and/or more adhesive traps than precut sheets to accommodate the reuse of assembled canopies.

(28) The embodiments discussed herein are illustrative of the present invention. As these embodiments of the present invention are described with reference to illustrations, various modifications or adaptations of the methods and or specific structures described may become apparent to those skilled in the art. All such modifications, adaptations, or variations that rely upon the teachings of the present invention, and through which these teachings have advanced the art, are considered to be within the spirit and scope of the present invention. Hence, these descriptions and drawings should not be considered in a limiting sense, as it is understood that the present invention is in no way limited to only the embodiments illustrated.

## Claims

1. An insect monitoring system comprising: a canopy including two parallel opposing vertical sides and a cover spanning between the opposing sides, the cover having two portions that subtend an

angle, wherein each opposing vertical side includes an angled slit and a hole, each angled slit extending from a bottom edge of the respective vertical side to the hole; an adhesive trap held vertically between the two opposing sides of the canopy; a lure held horizontally between the two portions of the cover and above a top edge of the adhesive trap; and means for hanging the insect monitoring system, the means attaching to the canopy.

2. The insect monitoring system of claim 1 wherein the cover includes a third portion between the two portions, the third portion being disposed horizontally and parallel to the lure.

3. The insect monitoring system of claim 1 wherein the two portions of the cover each include a horizontal slit sized to receive the lure therethrough.

4. The insect monitoring system of claim 1 wherein the opposing vertical sides each includes a tab, the two portions of the cover each includes slits, and each tab extends through one of the slits.

5. The insect monitoring system of claim 1 wherein each opposing vertical side includes a notch aligned with the hole and extending from the bottom edge towards the hole.

6. The insect monitoring system of claim 1 wherein the opposing vertical sides and the cover of the canopy comprise a single folded sheet of a rigid material.

7. The insect monitoring system of claim 1 wherein the means for hanging the insect monitoring system includes a wire.

8. The insect monitoring system of claim 7 wherein the cover includes two holes therethrough and the wire is disposed through the two holes.

9. The insect monitoring system of claim 8 wherein the lure includes a hole disposed therethrough, the hole in the lure is aligned with one of the two holes in the cover, and the wire is also disposed through the hole in the lure.

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